

Learning objectives

In this lesson you will learn to:

- Define what is meant by the term 'robust software'
- Explain how a hacker can exploit a code vulnerability
- Describe examples of bad coding practices and secure coding practices
- Explain how code reviews and audit trails help to identify vulnerabilities.

For more information and additional learner activities see Chapter 3.2 of the student book.

Robust software

You have considered how to make your own code robust in previous lessons.

Some of the problematic issues that might occur when a program is run are easy to predict – user errors, hardware faults, missing peripherals, etc.

Other potential issues are not so easy to predict, so programmers must make their code robust.

Robust software is developed so that it can handle all kinds of errors without crashing.

Vulnerabilities

Vulnerabilities are flaws in code that introduce security issues into software.

Hackers use these vulnerabilities to break into computer systems.

Some programming languages contain known vulnerabilities. For example, programs written in C access computer memory directly. This allows hackers to implement a 'back door' into systems.

Bad practice

Many software projects contain vulnerabilities that have been introduced due to bad programming practices. Such practices include:

- poorly planned design
- quick fixes for problems
- not using industry coding standards
- poorly structured code
- insufficient testing

Fixing bad practice

Code reviews can reduce some of these vulnerabilities. Automated reviews can be used to ensure programmers adhere to consistent standards. Programmers can review each other's code to ensure it is compliant and doesn't have any flaws.

Audit trails improve accountability by keeping track of who made what changes (and when) during the development process. Where necessary, this enables developers to restore a piece of code to the point it was at before a flaw was introduced.

Wrap up: you have learned how to...

- ✓ Define what is meant by the term 'robust software'.
 - Software that is designed to handle untoward occurrences without crashing or producing unexpected output.
- ✓ Explain how a hacker can exploit a code vulnerability.
 - Vulnerabilities introduce security flaws.
 - Hackers are able to use these flaws to break into computer systems. They can also insert their own code ('back doors') to do this.
- ✓ Describe examples of bad coding practices and secure coding practices.
 - Poor planning and testing, quick fixes and poor structure.
- ✓ Explain how code reviews and audit trails help to identify vulnerabilities.
 - Allow flaws to be monitored and tracked back to where they were introduced.
 - Reviewing code can catch vulnerabilities before a program goes live.